



Machine Learning

AIML CZG565

M5 : Decision Tree Classifier

BITS Pilani
Pilani Campus

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, Prof.Rajavadhana, Prof.Monali, Prof.Anita, Prof.Sangeetha, Prof.Swarna, Prof.Srinath

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

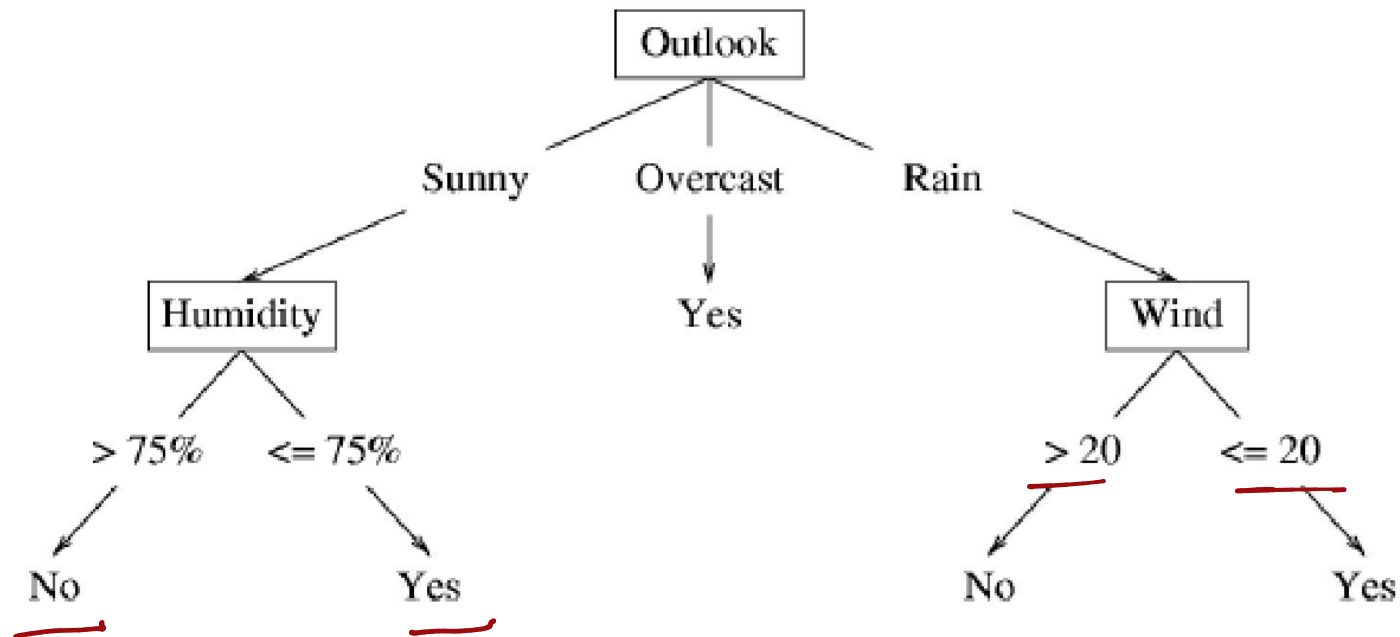
Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Agenda

- Information Theory
- Entropy Based Decision Tree Construction ✓
- Avoiding Overfitting
- Minimum Description Length
- Handling Continuous valued attributes

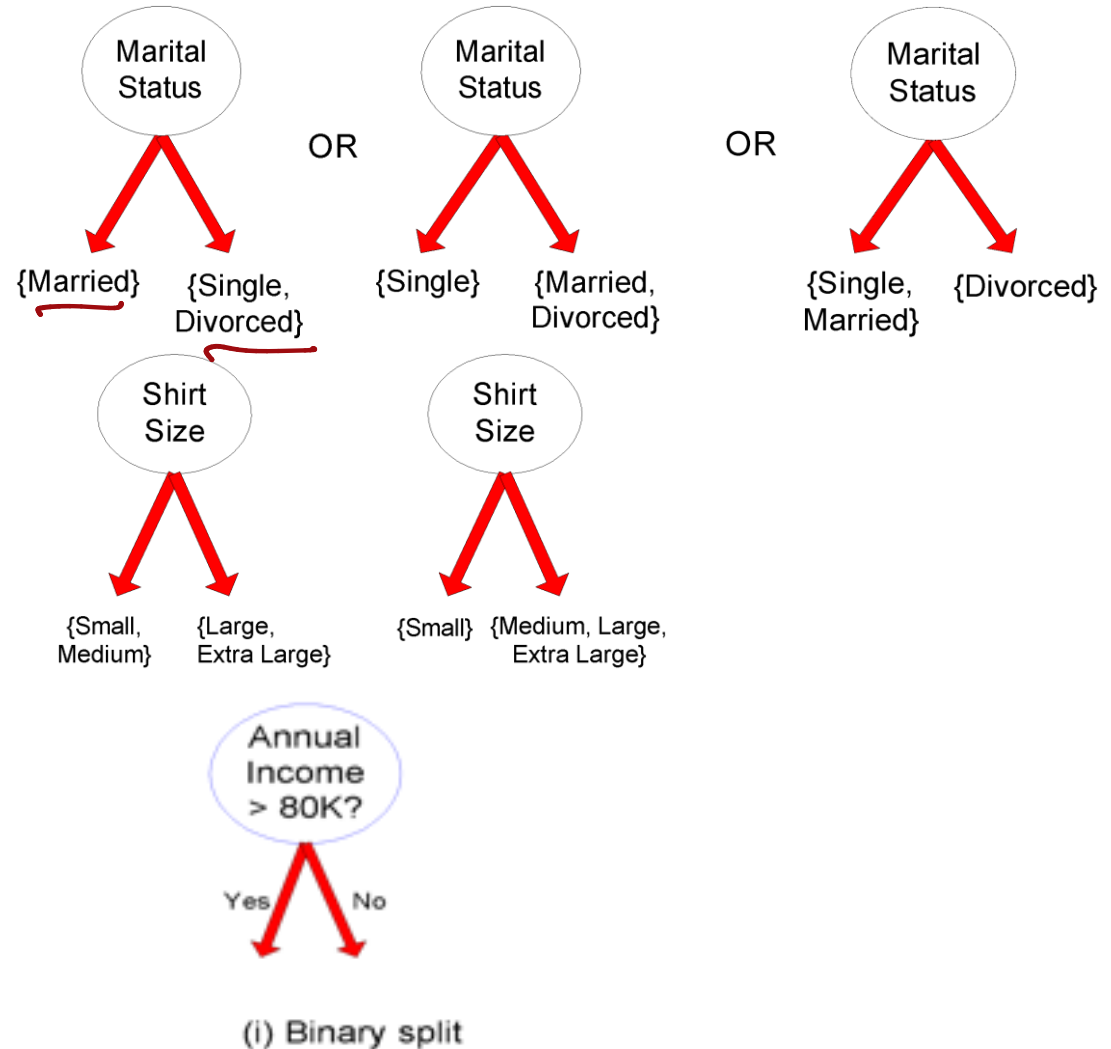
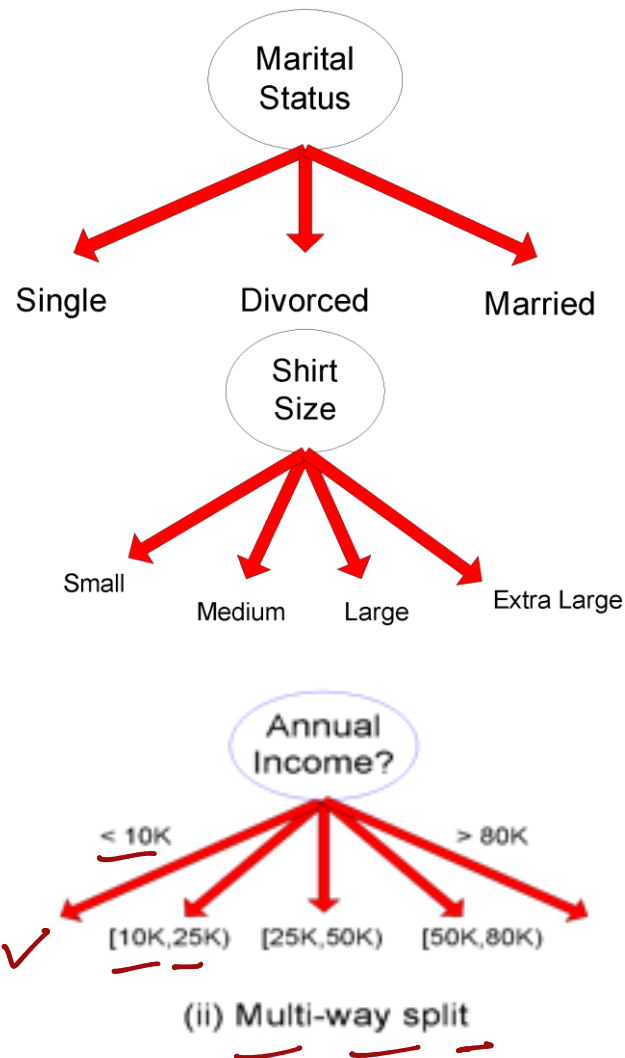
Decision Trees – Dealing with Continuous Values



Summary



Sample splits of Different Attribute Type



Decision Trees – Dealing with Continuous Values – Problem Type 5



- Given a continuous-valued attribute A , dynamically create a new attribute A_c

$A_c = \text{True if } A < c, \text{ False otherwise}$

- How to determine threshold value c ?

Way 1: Multi-way Split

- Example. *Temperature* in the *PlayTennis* example

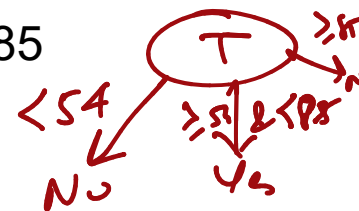
$$\frac{48 + 60}{2} = 54$$

$$\frac{80 + 90}{2} = 85$$

- Sort the examples according to *Temperature*

A	<i>Temperature</i>	40	<u>48</u>		<u>60</u>	<u>72</u>		<u>80</u>		<u>90</u>	
C/T	<i>PlayTennis</i>	No	No		54	Yes	Yes	Yes	(85)	No	

- Determine candidate thresholds by averaging consecutive values where there is a change in classification: $(48+60)/2=54$ and $(80+90)/2=85$



Decision Trees – Dealing with Continuous Values



- Given a continuous-valued attribute A , dynamically create a new attribute A_c

$A_c = \text{True if } A < c, \text{ False otherwise}$

- How to determine threshold value c ?

Way 2: Binary Splits

Best possible split

- Example. *Annual Income*

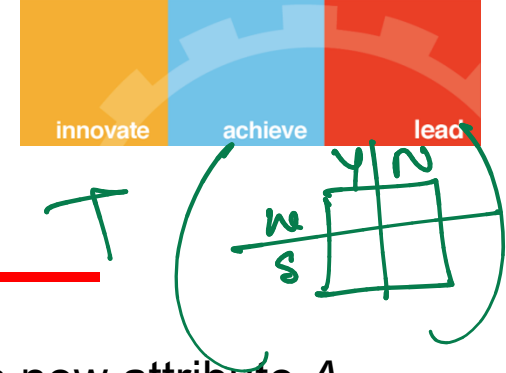
Sort the examples

according to *Annual Income*

Linearly scan all possible threshold, to determine best split point where the impurity is the least

ID	Home Owner	Marital Status	Annual Income	Defaulted?
1	Yes	Single	125000	No
2	No	Married	100000	No
3	No	Single	70000	No
4	Yes	Married	120000	No
5	No	Divorced	95000	Yes
6	No	Single	60000	No
7	Yes	Divorced	220000	No
8	No	Single	85000	Yes
9	No	Married	75000	No
10	No	Single	90000	Yes

Decision Trees – Dealing with Continuous Values – Problem Type 6



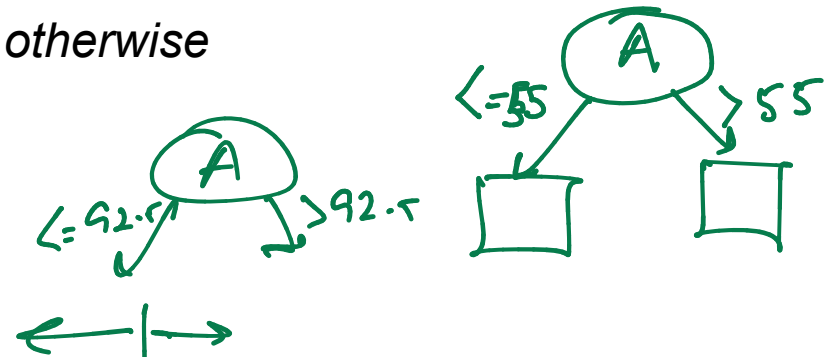
- Given a continuous-valued attribute A , dynamically create a new attribute A_c

$$A_c = \text{True if } A < c, \text{ False otherwise}$$

- How to determine threshold value c ?

Way 2: Binary Splits

- Example. *Annual Income*



Example: Annual Income

60 + 70 / 2

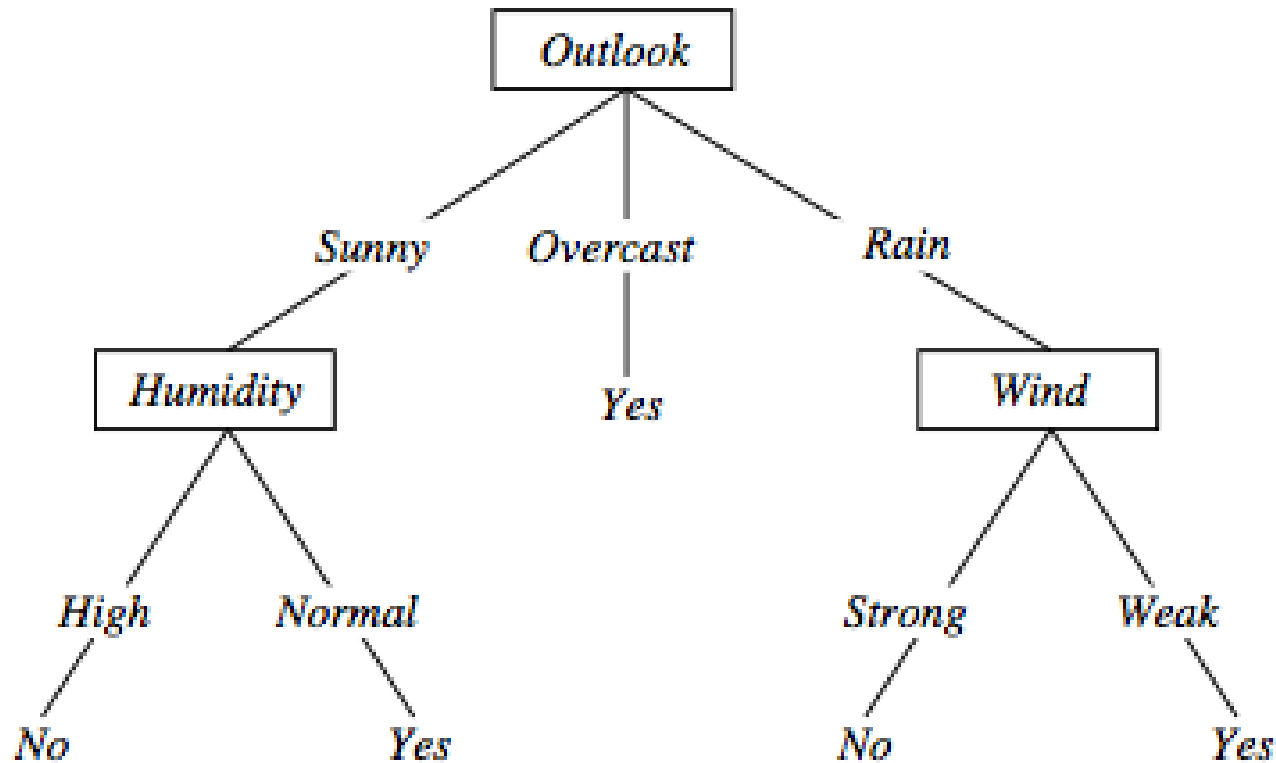
Sorted Values

Split Positions

lowest Entropy

Class	No	No	No	Yes	Yes	Yes	No	No	No	No		
Annual Income (in '000s)												
	60	70	75	85	90	95	100	120	125	220		
	55	65	72.5	80	87.5	92.5	97.5	110	122.5	172.5	230	
	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
Yes	0	3	0	3	0	3	1	2	2	1	3	0
No	0	7	1	6	2	5	3	4	3	4	3	4
Gini	0.420	0.400	0.375	0.343	0.417	0.400	0.300	0.343	0.375	0.400	0.420	

Overfitting in decision trees

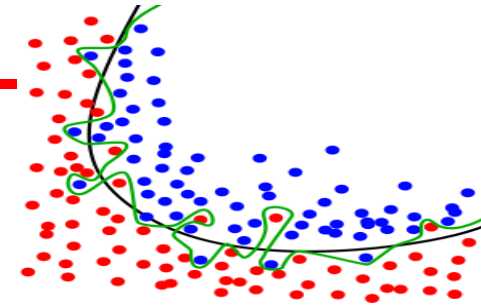


$\langle \text{Outlook}=\text{Sunny}, \text{Temp}=\text{Hot}, \text{Humidity}=\text{Normal}, \text{Wind}=\text{Strong}, \text{PlayTennis}=\text{No} \rangle$

New noisy example causes splitting of second leaf node.

Overfitting and Tree Pruning

- Overfitting: An induced tree may overfit the training data
 - Too many branches, may reflect anomalies due to noise or outliers in the training data
 - Poor accuracy for unseen samples



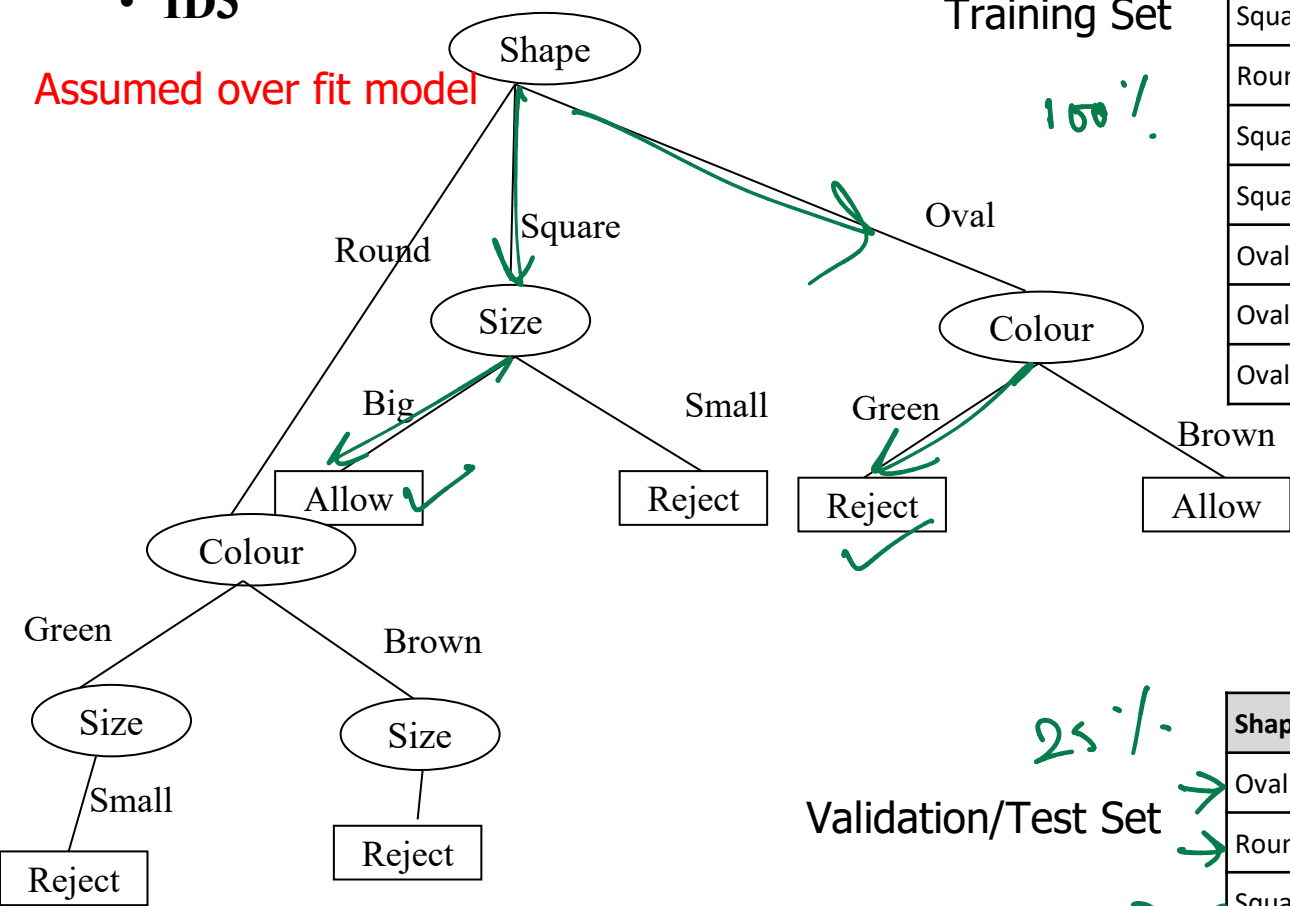
Approaches to tackle:

- ✓ Pruning
- Constraint on the depth of tree
- Constraint on the minimum nodes allowed at leaf

Over fitting

• ID3

Assumed over fit model



Training Set

100%

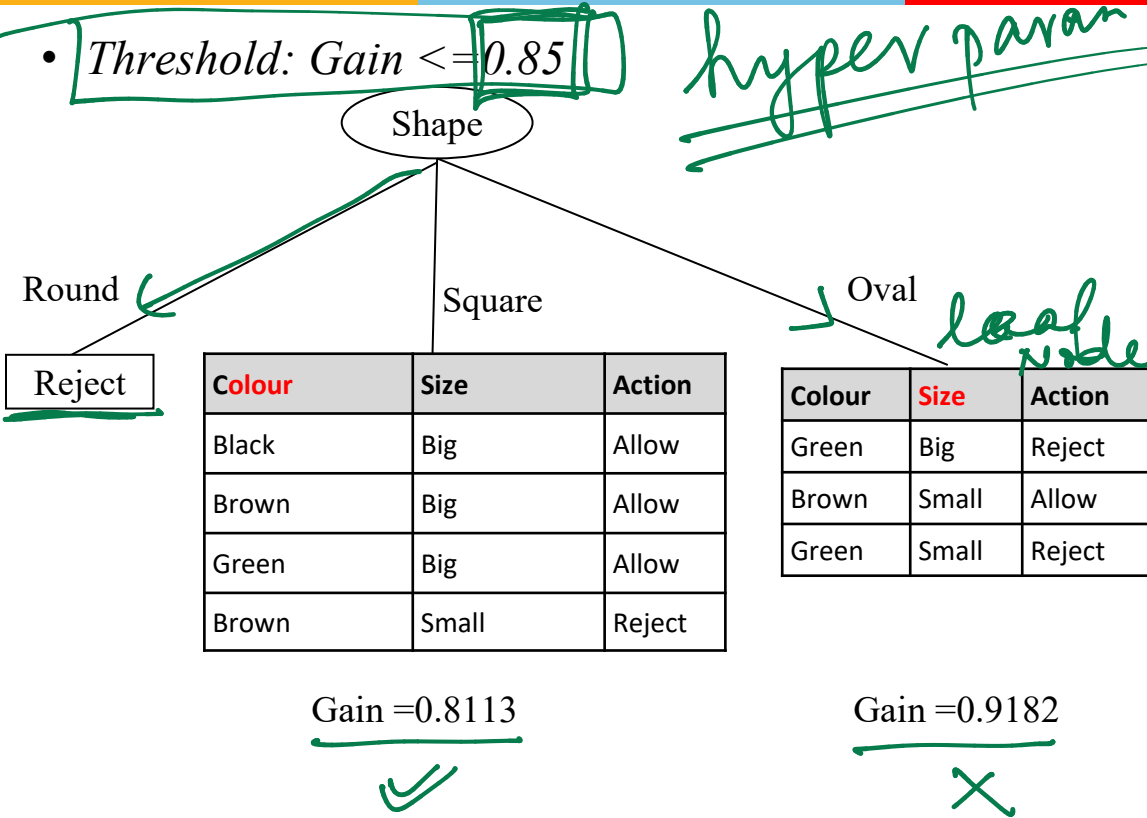
Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

Validation/Test Set

25%

Shape	Colour	Size	Action
Oval	Black	Small	Reject
Round	Brown	Big	Allow
Square	Brown	Big	Allow
Oval	Green	Small	Allow

Pre-Pruning – Problem Type 7

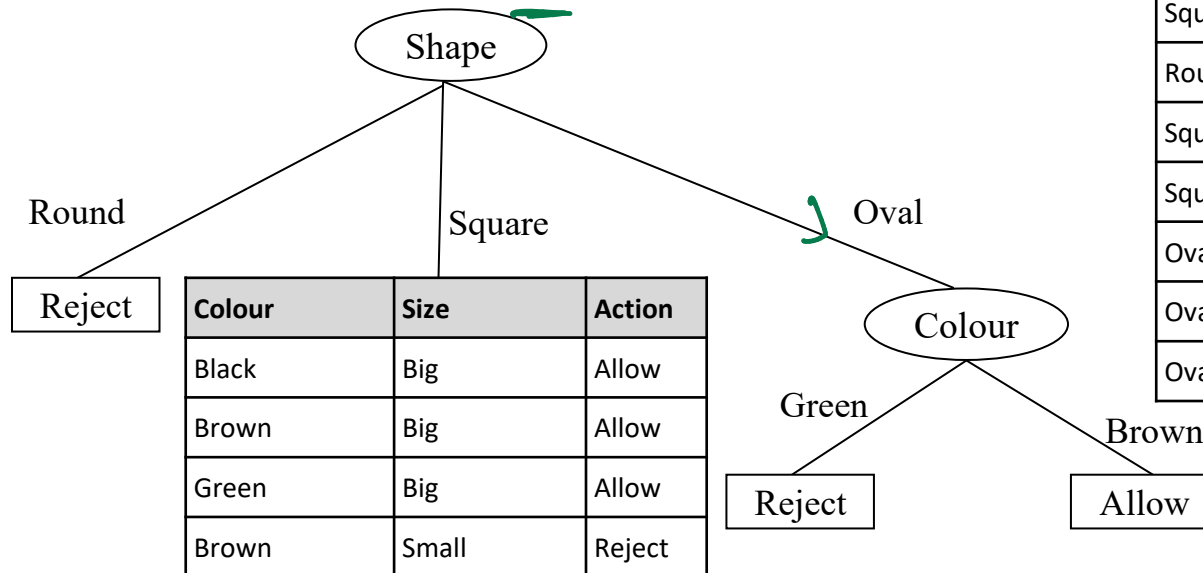


Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

Shape	Colour	Size	Action
Oval	Black	Small	Reject
Round	Brown	Big	Allow
Square	Brown	Big	Allow
Oval	Green	Small	Allow

Pre-Pruning – Problem Type 7

- Threshold: $Gain \leq 0.85$



Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

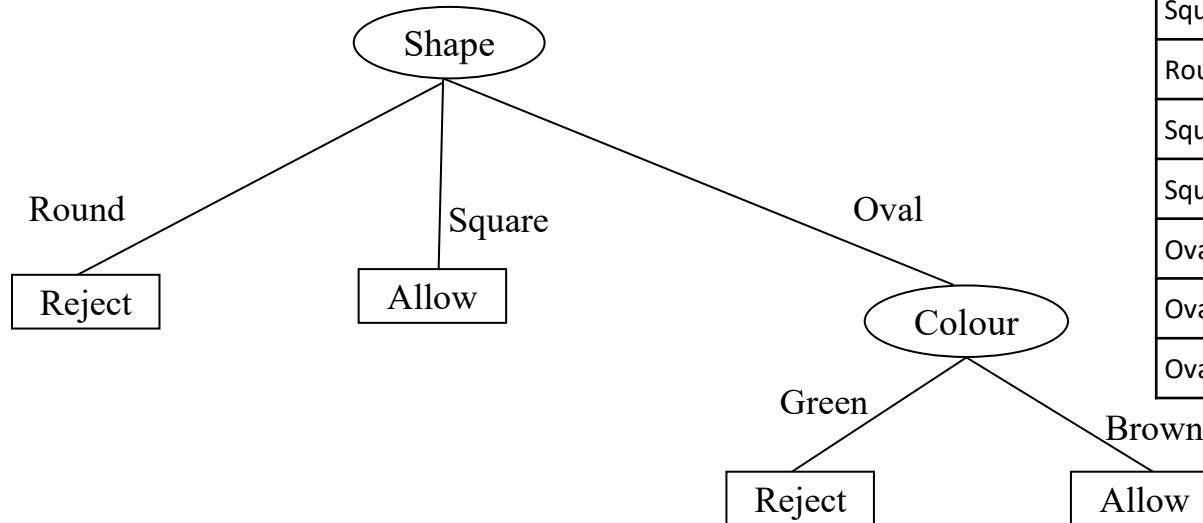
Note:

- DT can estimate the probability of instance's membership to a class

Shape	Colour	Size	Action
Oval	Black	Small	Reject
Round	Brown	Big	Allow
Square	Brown	Big	Allow
Oval	Green	Small	Allow

Pre-Pruning – Problem Type 7

- Threshold: $Gain \leq 0.85$



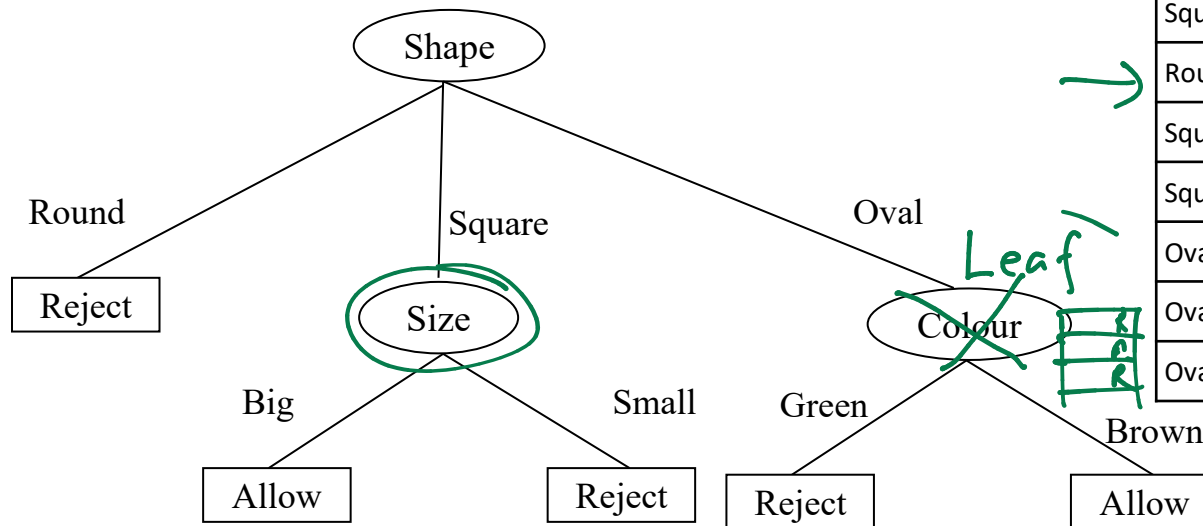
Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

Apply majority voting for converting the pruned subset of data into a class

Shape	Colour	Size	Action
Oval	Black	Small	Reject
Round	Brown	Big	Allow
Square	Brown	Big	Allow
Oval	Green	Small	Allow

Post-Pruning – Problem Type 8

Assumed model from the output of one intermediate iteration of the decision tree building



Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

Idea :

1. Post construction, scan the tree bottom-up
2. At every decision node

Retain the attribute node & evaluate it against the prune set (validation set)

Remove the attribute node & reevaluate it with the same prune set

If there is a reduction in error, prune the node else retain the node

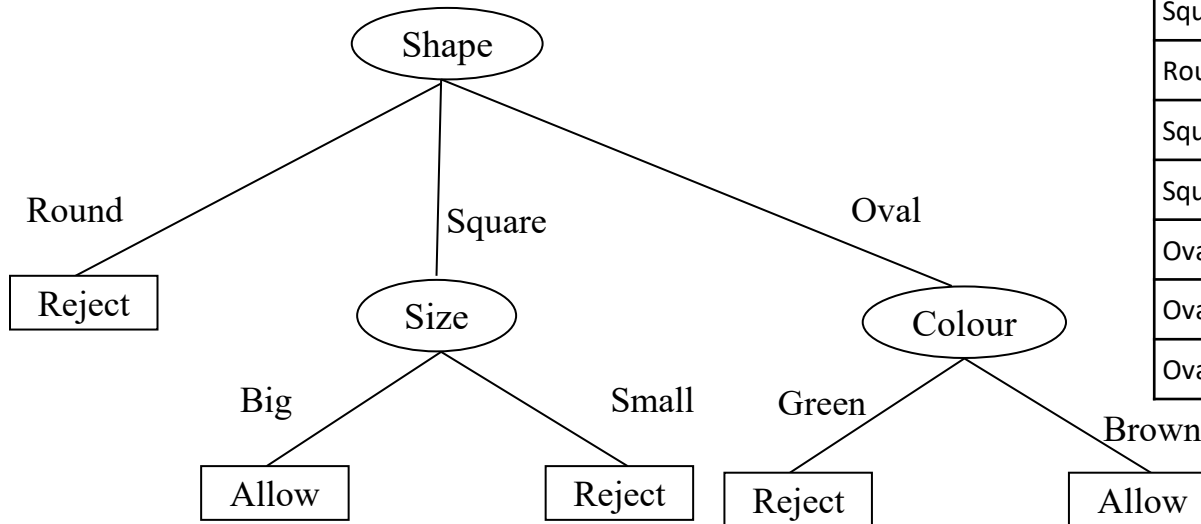
3. Repeat this in other branches of the tree

Prune Set

Shape	Colour	Size	Action
Oval	Black	Small	Reject
Round	Brown	Big	Allow
Square	Brown	Big	Allow
Oval	Green	Small	Allow

Post-Pruning – Problem Type 8

Assumed model from the output of one intermediate iteration of the decision tree building



Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

- Error rate is the percentage of tuples misclassified
- Prune set is used to estimate the cost

Prune Set□

Shape	Colour	Size	Action
Oval	Black	Small	Reject
Round	Brown	Big	Allow
Square	Brown	Big	Allow
Oval	Green	Small	Allow

Post-Pruning – Problem Type 8

Shape

Round

Reject

Square

Size

Big

Allow

Small

Reject

Oval

Colour

Green

Reject

Brown/Others

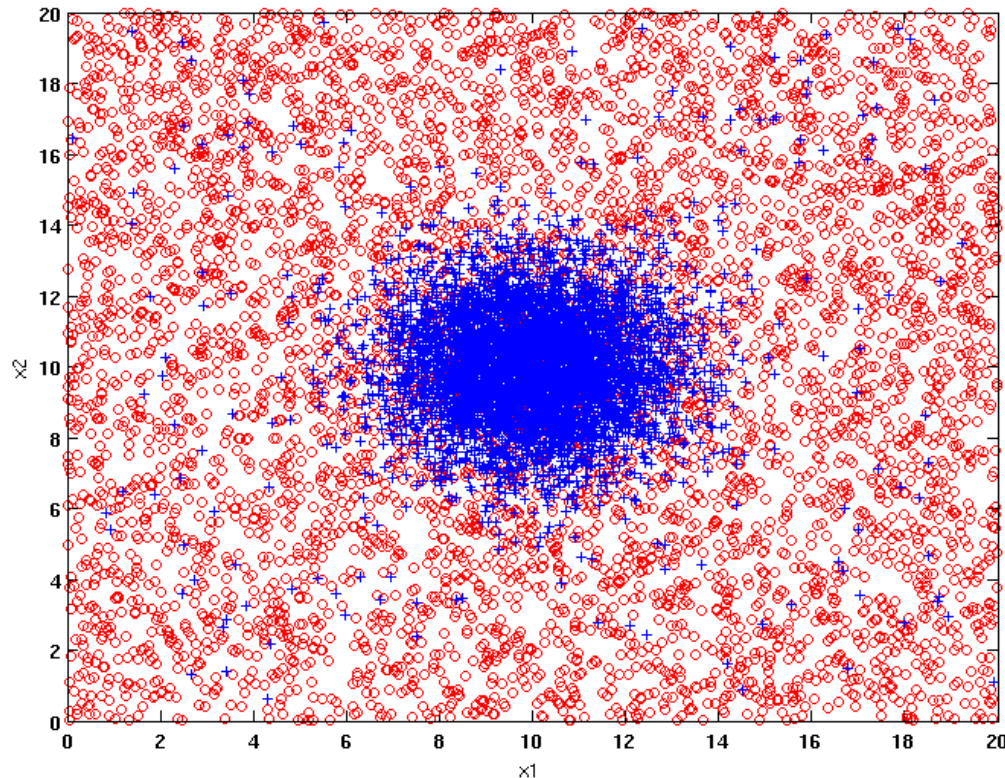
Allow

Shape	Colour	Size	Action
Round	Green	Small	Reject
Square	Black	Big	Allow
Square	Brown	Big	Allow
Round	Brown	Small	Reject
Square	Green	Big	Allow
Square	Brown	Small	Reject
Oval	Green	Big	Reject
Oval	Brown	Small	Allow
Oval	Green	Small	Reject

Prune Size & Predict	Prune Colour & Predict	Above Tree's Prediction	Shape	Colour	Size	Action
Allow	Reject ✓	Allow	Oval	Black	Small	Reject ✗
Reject	Reject ✗	Reject	Round	Brown	Big	Allow ✗
Allow	Allow ✓	Allow	Square	Brown	Big	Allow ✓
Reject	Reject ✗	Reject	Oval	Green	Small	Allow ✗

Example Data Set

Studying Overfitting & Bias–Variance Tradeoff



Two class problem:

+ : 5200 instances

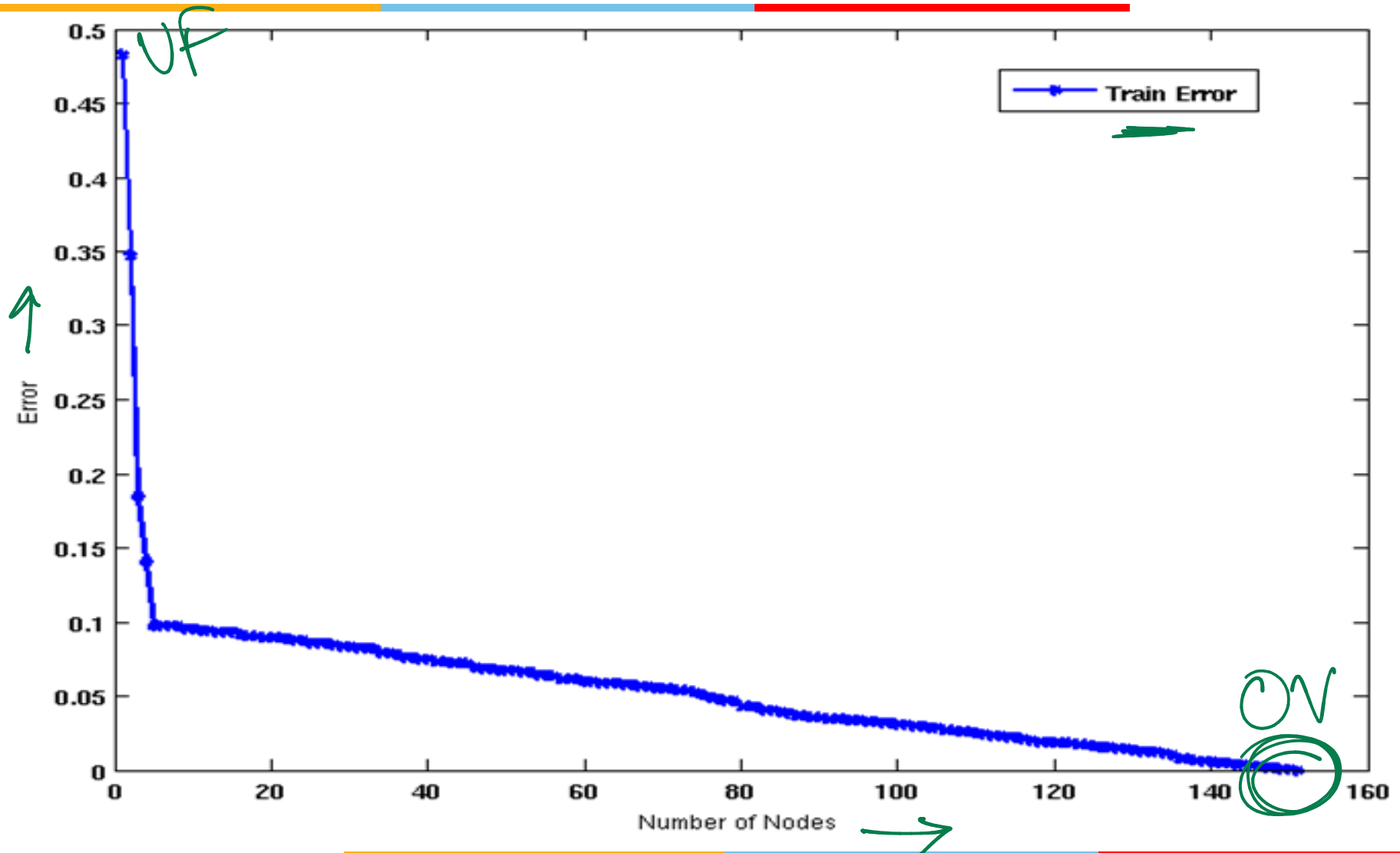
- 5000 instances generated from a Gaussian centered at (10,10)
- 200 noisy instances added

o : 5200 instances

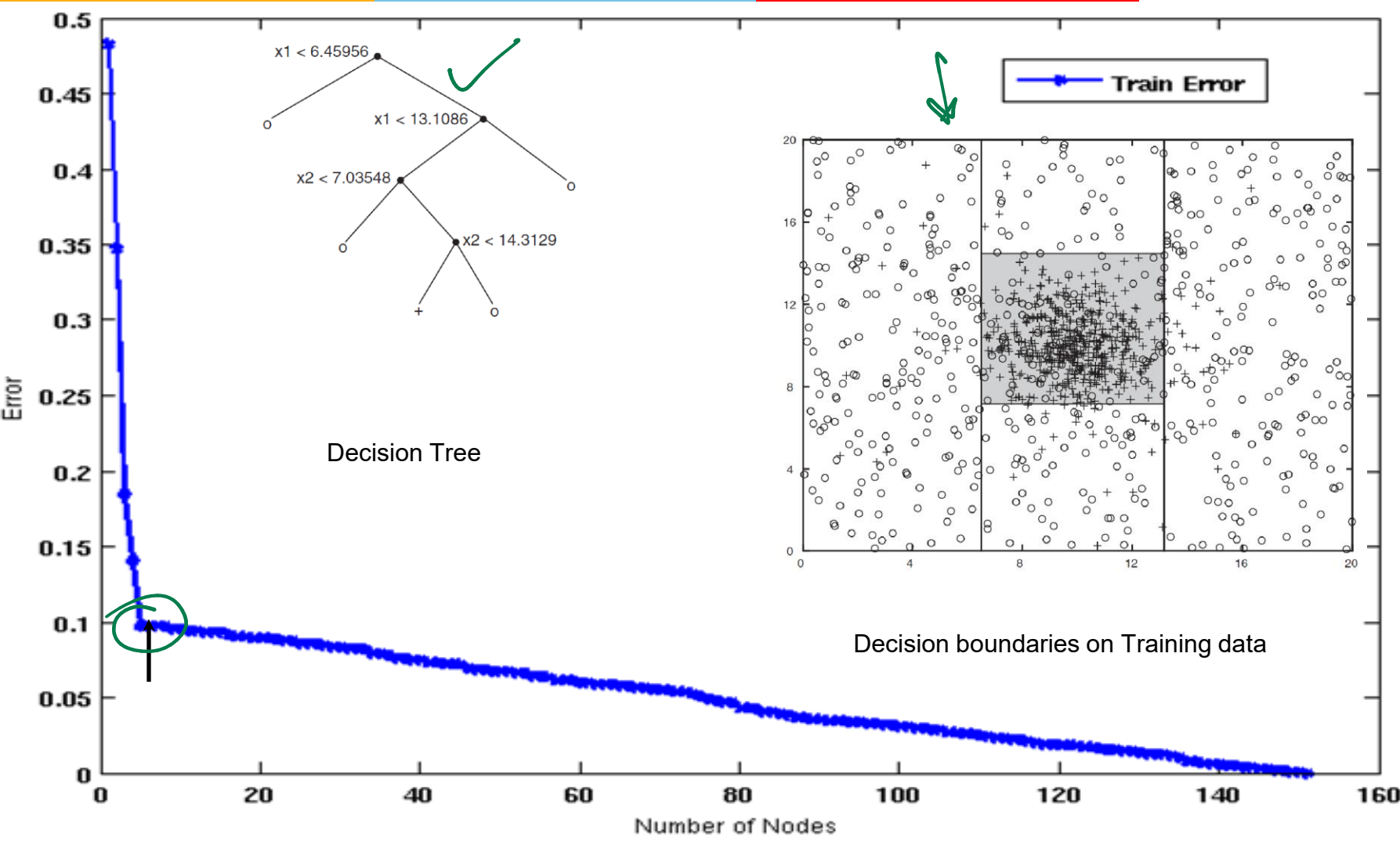
- Generated from a uniform distribution

10 % of the data used for training and 90% of the data used for testing

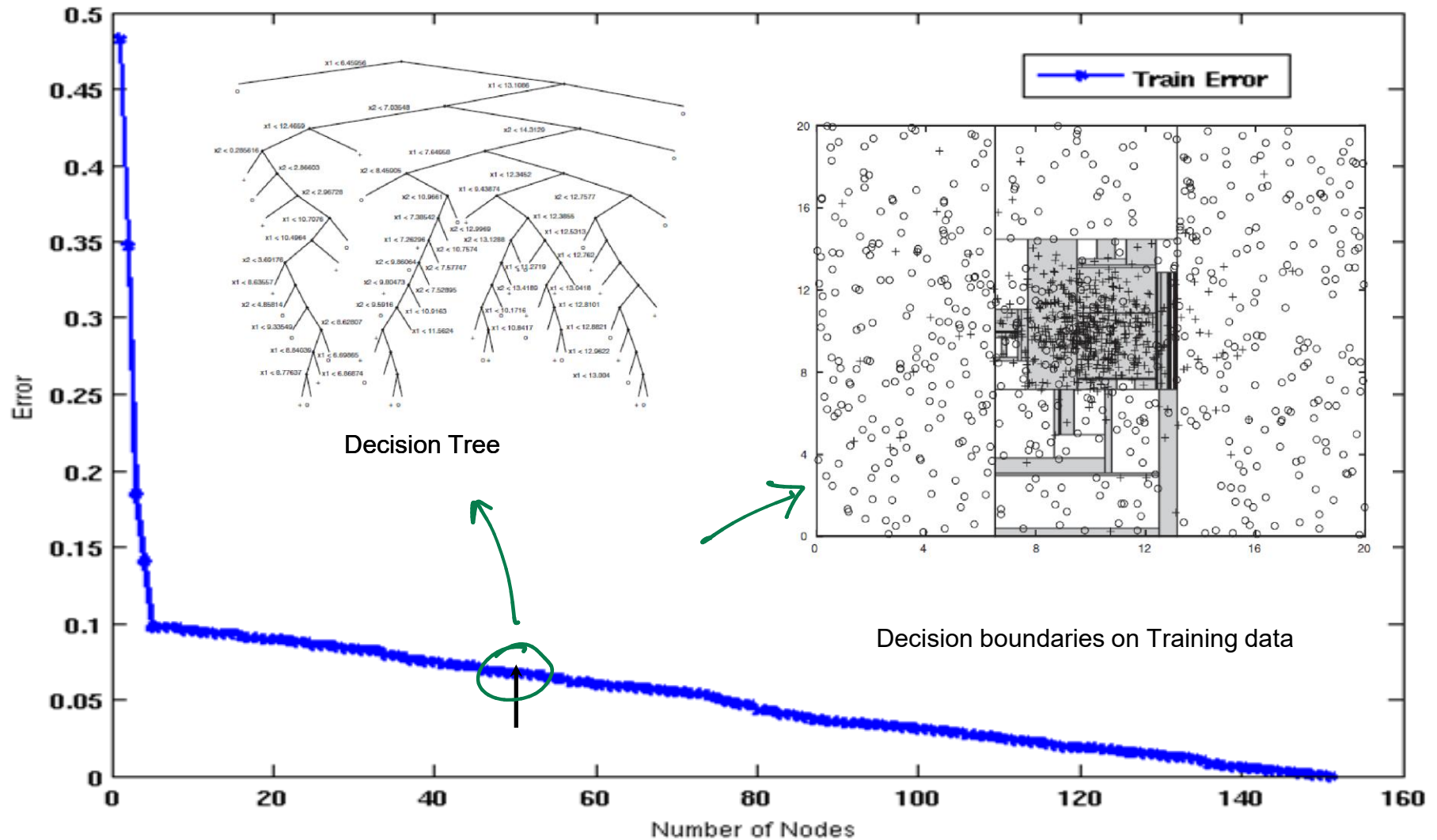
Increasing number of nodes in Decision Trees



Decision Tree with 4 nodes

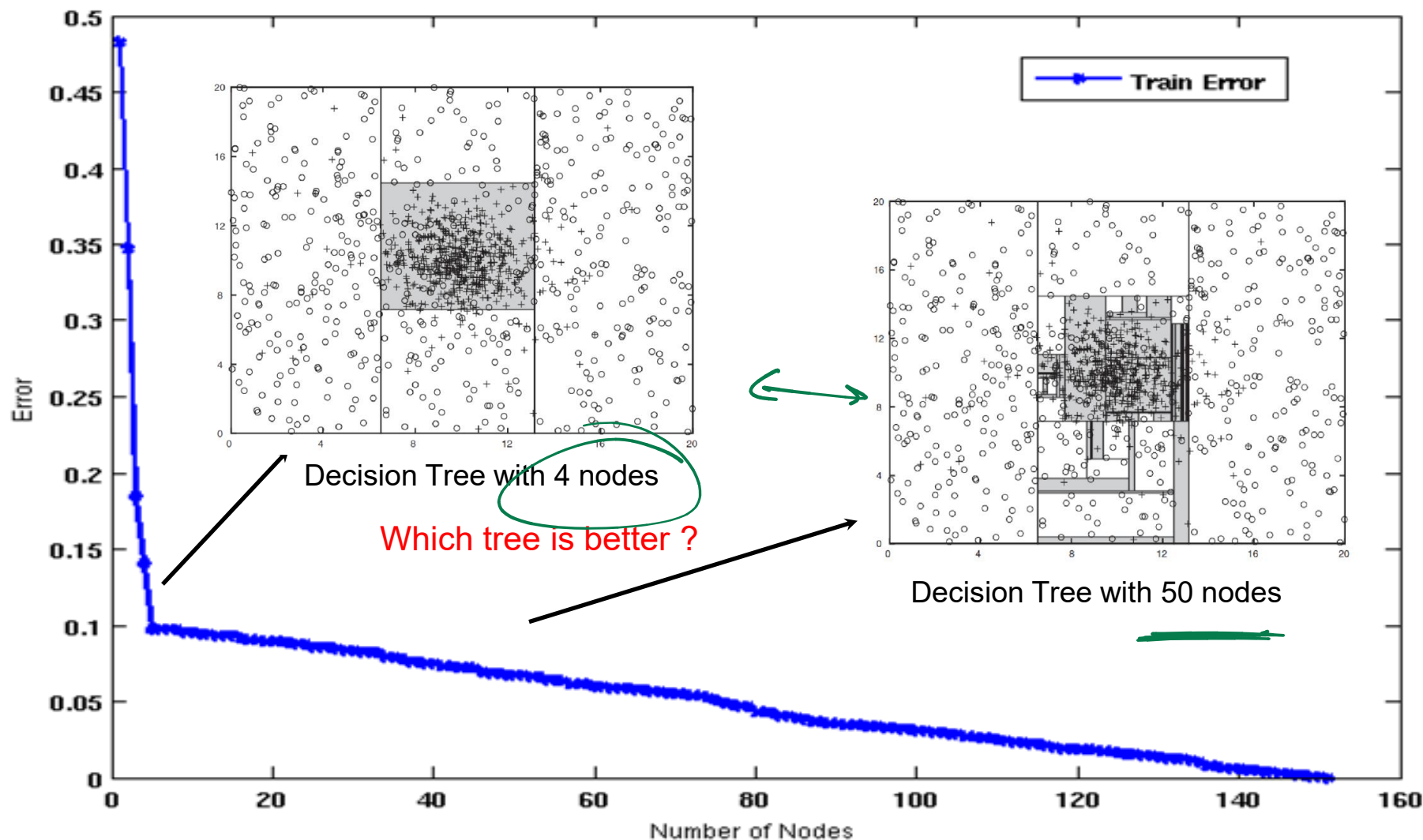


Decision Tree with 50 nodes

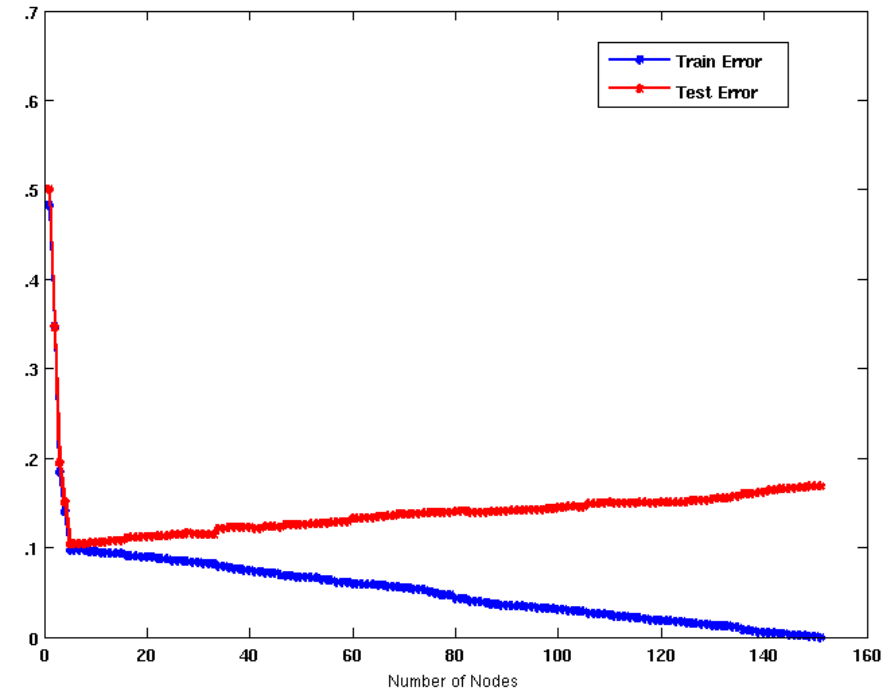
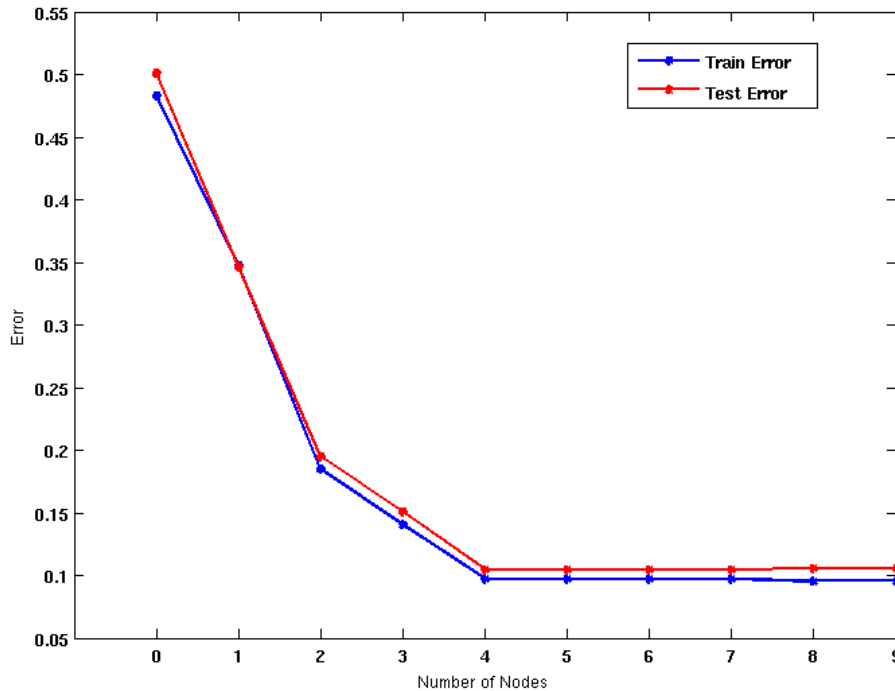


Which tree is better?

Problem Type 9 – Model Comparison



Model Overfitting



Underfitting: when model is too simple, both training and test errors are large

Overfitting: when model is too complex, training error is small but test error is large

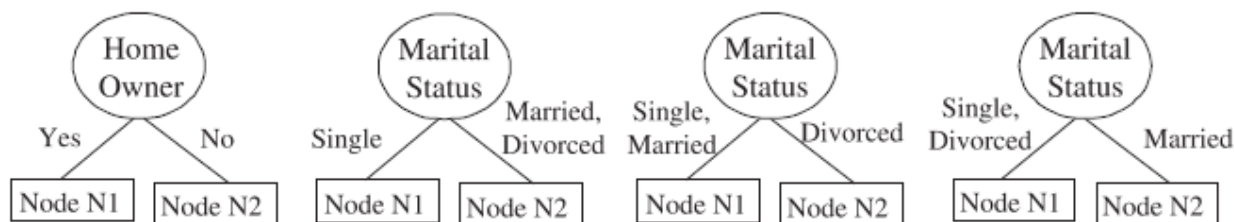
Alternative to Entropy:

Gini Index : Binary Splits



$$H(x) = - \sum_{i=1}^n p_i \log p_i$$

ID	Home Owner	Marital Status	Annual Income	Defaulted?
1	Yes	Single	125000	No
2	No	Married	100000	No
3	No	Single	70000	No
4	Yes	Married	120000	No
5	No	Divorced	95000	Yes
6	No	Single	60000	No
7	Yes	Divorced	220000	No
8	No	Single	85000	Yes
9	No	Married	75000	No
10	No	Single	90000	Yes



Alternative to Entropy:

Gini Index : Binary Splits



Gini Index: Parent

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

ID	Home Owner	Marital Status	Annual Income	Defaulted?
1	Yes	Single	125000	No
2	No	Married	100000	No
3	No	Single	70000	No
4	Yes	Married	120000	No
5	No	Divorced	95000	Yes
6	No	Single	60000	No
7	Yes	Divorced	220000	No
8	No	Single	85000	Yes
9	No	Married	75000	No
10	No	Single	90000	Yes

$$1 - \left[\left(\frac{3}{10} \right)^2 + \left(\frac{7}{10} \right)^2 \right] = 1 - \left(\frac{3}{10} \right)^2 - \left(\frac{7}{10} \right)^2 = 0.420.$$

$\xrightarrow{p(\text{Yes})}$ $\xrightarrow{p(\text{No})}$

	<u>Parent</u>
No	7
Yes	3
Gini = 0.420	

$$1 - \sum_i p_i^2$$

faster than log.

Alternative to Entropy:

Gini Index : Binary Splits



Splitting Attribute: HomeOwner

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Gini index for the child nodes

$$N1 = 1 - (0/3)^2 - (3/3)^2 = 0$$

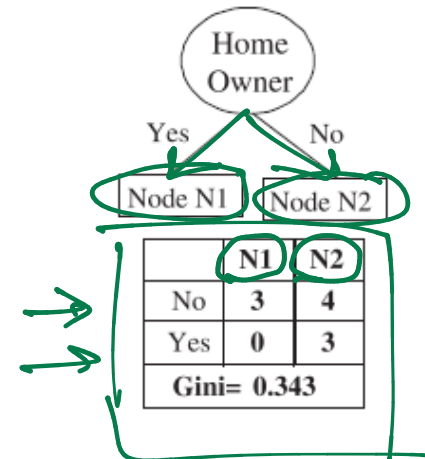
$$N2 = 1 - (3/7)^2 - (4/7)^2 = 0.490$$

Weighted Gini Index for children

$$I(\text{children}) = \sum_{j=1}^k \frac{N(v_j)}{N} I(v_j)$$

$$(3/10) \times 0 + (7/10) \times 0.490 = 0.343$$

ID	Home Owner	Marital Status	Annual Income	Defaulted?
1	Yes	Single	125000	No
2	No	Married	100000	No
3	No	Single	70000	No
4	Yes	Married	120000	No
5	No	Divorced	95000	Yes
6	No	Single	60000	No
7	Yes	Divorced	220000	No
8	No	Single	85000	Yes
9	No	Married	75000	No
10	No	Single	90000	Yes



The gain using Home Owner as splitting attribute is
 $0.420 - 0.343 = 0.077$

Alternative to Entropy:

Gini Index : Binary Splits

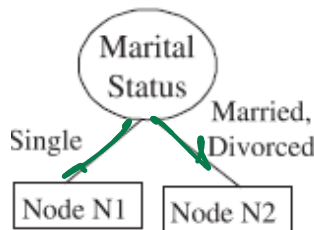
innovate

achieve

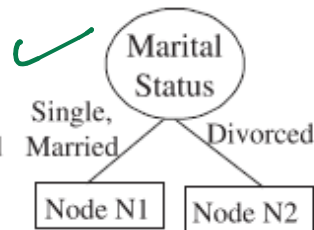
lead

Splitting Attribute: MaritalStatus

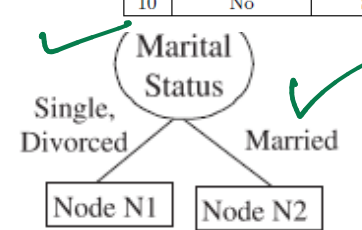
ID	Home Owner	Marital Status	Annual Income	Defaulted?
1	Yes	Single	125000	No
2	No	Married	100000	No
3	No	Single	70000	No
4	Yes	Married	120000	No
5	No	Divorced	95000	Yes
6	No	Single	60000	No
7	Yes	Divorced	220000	No
8	No	Single	85000	Yes
9	No	Married	75000	No
10	No	Single	90000	Yes



	N1	N2
No	3	4
Yes	2	1
Gini= 0.400		



	N1	N2
No	6	1
Yes	2	1
Gini= 0.400		



	N1	N2
No	4	3
Yes	3	0
Gini= 0.343		

Based on these results, Home Owner and the last binary split using Marital Status are clearly the best candidates, since they both produce the lowest weighted average Gini index

An alternative measure: gain ratio

$$SplitInformation(S, A) \equiv - \sum_{i=1}^c \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}$$

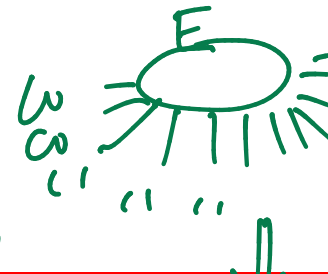
- S_i are the sets obtained by partitioning on value i of A
- *SplitInformation* measures the entropy of S with respect to the values of A . The more uniformly dispersed the data the higher it is.

$$GainRatio(S, A) \equiv \frac{Gain(S, A)}{SplitInformation(S, A)}$$

- *GainRatio* penalizes attributes that split examples in many small classes such as *Date*. Let $|S| = n$, *Date* splits examples in n classes
 - $SplitInformation(S, Date) = -[(1/n \log_2 1/n) + \dots + (1/n \log_2 1/n)] = -\log_2 1/n = \log_2 n$
- Compare with A , which splits data in two even classes:
 - $SplitInformation(S, A) = -[(1/2 \log_2 1/2) + (1/2 \log_2 1/2)] = -[-1/2 - 1/2] = 1$

Limitation of Information Gain:

Ways to tackle



SplitInformation measures the entropy of S with respect to the values of A. The more uniformly dispersed the data the higher it is.

less ↓ high ↑

$$\text{SplitInformation}(S, A) \equiv - \sum_{i=1}^c \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}$$

$$\text{SplitInformation}(S, \text{customerID}) = -[(1/n \log_2 1/n) + \dots + (1/n \log_2 1/n)] = -\log_2 1/n = \log_2 n$$

GainRatio penalizes attributes that split examples in many small classes such as ID by incorporating split information.

$$\text{GainRatio}(S, A) \equiv \frac{\text{Gain}(S, A)}{\text{SplitInformation}(S, A)}$$

high

high

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

Select the best attribute test condition among the following three attributes:
Gender, Car Type, and Customer ID

Limitation of Information Gain:

Ways to tackle

Problem Type 10

$$\rightarrow \text{Entropy}(\text{parent}) = -\frac{10}{20} \log_2 \frac{10}{20} - \frac{10}{20} \log_2 \frac{10}{20} = 1.$$

If Gender is used as attribute test condition:

$$\rightarrow \text{Entropy}(\text{children}) = \frac{10}{20} \left[-\frac{6}{10} \log_2 \frac{6}{10} - \frac{4}{10} \log_2 \frac{4}{10} \right] \times 2 = 0.971$$

$$\text{Gain Ratio} = \frac{1 - 0.971}{-\frac{10}{20} \log_2 \frac{10}{20} - \frac{10}{20} \log_2 \frac{10}{20}} = \frac{0.029}{1} = 0.029$$

-Σ p log p

If Car Type is used as attribute test condition:

$$\text{Entropy}(\text{children}) = \frac{4}{20} \left[-\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} \right] + \frac{8}{20} \times 0 + \frac{8}{20} \left[-\frac{1}{8} \log_2 \frac{1}{8} - \frac{7}{8} \log_2 \frac{7}{8} \right] = 0.380$$

$$\text{Gain Ratio} = \frac{1 - 0.380}{-\frac{4}{20} \log_2 \frac{4}{20} - \frac{8}{20} \log_2 \frac{8}{20} - \frac{8}{20} \log_2 \frac{8}{20}} = \frac{0.620}{1.52} = 0.41$$

↓

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

Limitation of Information Gain:

Ways to tackle

Problem Type 10



Finally, if Customer ID is used as attribute test condition:

$$\text{Entropy(children)} = \frac{1}{20} \left[-\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \right] \times 20 = 0$$

$$\text{Gain Ratio} = \frac{1 - 0}{-\frac{1}{20} \log_2 \frac{1}{20} \times 20} = \frac{1}{4.32} = 0.23$$

$$-\sum p \log p$$

splits

$$P_M = \frac{10}{20} \quad P_F = \frac{10}{20}$$



Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

Inference : Thus, even though Customer ID has the highest information gain, its gain ratio is lower than Car Type since it produces a larger number of splits

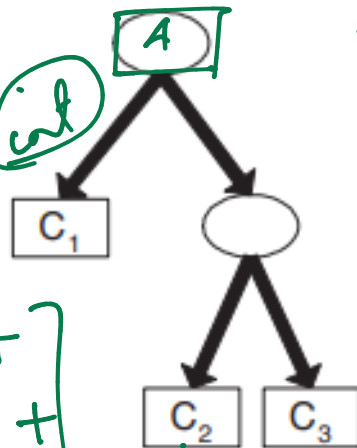
MDL

Minimum Description Length Principle

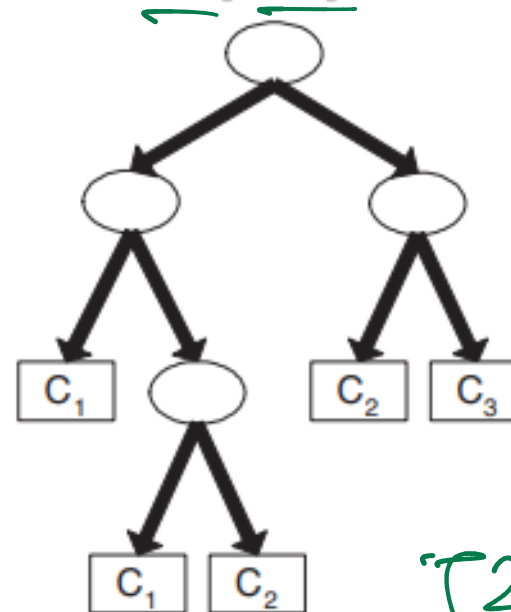
Problem Type 11

Consider the decision trees shown in Figure 1. Assume they are generated from a data set that contains 16 binary attributes and 3 classes, C_1 , C_2 , and C_3 . [Compute the total description length of each decision tree according to the minimum description length principle.]

Which decision tree is better, according to the MDL principle?



(a) Decision tree with 7 errors



(b) Decision tree with 4 errors

$$2 \rightarrow 1$$

$$3 \rightarrow 2$$

$$4 \rightarrow 2$$

$$\lceil \log_2 C \rceil$$

$$\text{int} = 4$$

$$\text{leaf} = 5$$

$$\left[4 \times 4 + 5 \times 2 + 4 \log_2^n \right]$$

$$n =$$

$$14 + 7 \log_2^n$$

$$26 + 4 \log_2^n$$

① NO CODING Qn For ML

② NO DIRECT Qn

③ Numerical ✓

④ Conceptual / Analysis ✓

Sample Paper
Taxila

R/M - same

Minimum Description Length Principle

Problem Type

11

Answer:

Because there are 16 attributes, the cost for each internal node in the decision tree is:

$$\log_2(m) = \log_2(16) = 4 \quad \checkmark$$

$$16 \rightarrow 4 \text{ bits}$$

Furthermore, because there are 3 classes, the cost for each leaf node is:

$$\lceil \log_2(k) \rceil = \lceil \log_2(3) \rceil = 2 \quad \checkmark$$

The cost for each misclassification error is $\log_2(n)$.

$$7 \rightarrow 7 \cdot \log_2 n$$

The overall cost for the decision tree (a) is $2 \times 4 + 3 \times 2 + 7 \times \log_2 n = 14 + 7 \log_2 n \quad \checkmark$

and the overall cost for the decision tree (b) is $4 \times 4 + 5 \times 2 + 4 \times 5 = 26 + 4 \log_2 n. \quad \checkmark$

According to the MDL principle, tree (a) is better than (b) if $n < 16$ and is worse than (b) if $n > 16$.

$$\begin{array}{ccc} n=8 & n=16 & n=32 \\ (a) & = & (b) \end{array}$$

$$\begin{array}{l}
 \left[\begin{array}{l}
 \# \text{int} \times C_{\text{int}} \xrightarrow{+} \lceil \log_2 \text{Attr} \rceil \text{ in binary} \\
 \# \text{leaf} \times C_{\text{leaf}} \xrightarrow{+} \lceil \log_2 \text{classes} \rceil \\
 \# \text{err} \times C_{\text{err}} \xrightarrow{\quad} \log_2 \text{data points}
 \end{array} \right.
 \end{array}$$

Minimum Description Length Principle

$$\begin{aligned}
 h_{MAP} &= \arg \max_{h \in H} P(D|h)P(h) \\
 &= \arg \max_{h \in H} \log_2 P(D|h) + \log_2 P(h) \quad \checkmark \\
 &= \arg \min_{h \in H} -\log_2 P(D|h) - \log_2 P(h) \quad (1) \quad \checkmark
 \end{aligned}$$

$P(h/D) \propto P(D/h)P(h)$

Interesting fact from information theory:

The optimal (shortest expected coding length) code for an event with probability p is $-\log_2 p$ bits.

So interpret (1):

- $L_{C1}(h) = \text{length}(h) = -\log_2 P(h)$ [cost to encode the model] → int + leaf
- $L_{C2}(D|h) = \text{length}(\text{misclassifications}) = -\log_2 P(D|h)$ [cost to encode the data errors given the model]

→ prefer the hypothesis that minimizes

$$\text{length}(h) + \text{length}(\text{misclassifications})$$

Minimum Description Length Principle

- MDL principle provides a way of trading off hypothesis complexity for the number of errors committed by the hypothesis.
- May select a shorter hypothesis that makes a few errors over a longer hypothesis that perfectly classifies the training data.
- Provides one method for dealing with the issue of overfitting the data.

Minimum Description Length Principle

Occam's razor: prefer the shortest hypothesis

MDL: prefer the hypothesis h that minimizes

$$h_{MDL} = \underset{h \in H}{\operatorname{argmin}} L_{C_1}(h) + L_{C_2}(D|h)$$

where $L_C(x)$ is the description length of x under encoding C

Example: H = decision trees hypothesis, D = training data labels

- $L_{C_1}(h)$ is # bits to describe tree h
- $L_{C_2}(D|h)$ is # bits to describe D given h
 - Note $L_{C_2}(D|h) = 0$ if examples classified perfectly by h . Need only describe exceptions
- Hence h_{MDL} trades off tree size for training errors

DT
log₂

~~C&L~~ loge (ln)

Practice Exercises (for Students)

- Which of the below discretization best suits to transform the attribute "Annual Income" for decision tree construction? Use entropy as decision criteria.
 - Annual Income ($\leq 85k$, $> 85k$ & $\leq 200k$, $> 200k$)
 - Annual Income ($\leq 90k$, $> 90k$)
- If binary split is recommended for attribute "Marital Status" which of the combination of splits best fits? Justify your comment using entropy.
- Use the results of part 1) & part 2) and Build a decision tree classifier using ID3 algorithm i.e., Information gain and entropy measures. Grow the complete decision tree.
- Use the test data to post prune the built tree. Try to prune atleast two internal nodes and choose the best of the trials

	binary	categorical	continuous	class
Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Test Data / Prune Set/ Validation Set			
Owner	Marital	Income	Default
No	Single	85k	No
Yes	Single	100k	Yes
Yes	Divorced	70k	No
No	Married	90k	Yes

Additional References

Decision Tree

- https://www.youtube.com/watch?v=eKD5gxPPeY0&list=PLBv09BD7ez_4temBw7vLA19p3tdQH6FYO&index=1

Overfitting

- https://www.youtube.com/watch?time_continue=1&v=t56Nid85Thg
- <https://www.youtube.com/watch?v=y6SpA2Wuyt8>

Random Forest

- <https://www.stat.berkeley.edu/~breiman/RandomForests/>

Thank you !